

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial #1 - yields and stoichiometry

Question 1

Bacteria is growing in a 5 cubic meter (m^3) batch fermenter in aerobic conditions. The nitrogen source in the fermentation is ammonia and the carbon source is glycerol ($MW=92 \frac{g}{mol}$, $C_3H_8O_3$) in initial concentration of $10 \frac{g}{lit}$. At the end of the fermentation, the glycerol concentration is $1 \frac{g}{lit}$. The products in the process are biomass, CO_2 and water only, and the given yield coefficient is $0.6 \text{ gr biomass/gr glycerol}$.

The empiric formula of the bacteria is $CH_{1.74}O_{0.43}N_{0.22}$ ($MW=23.7 \frac{gX}{mol}$).

A. What is the $Y_{O_2/S}$ yield (mole O_2 /mole Substrate)?

$$(1.04 \frac{\text{mole } O_2}{\text{mole Substrate}})$$

B. What is the total amount of heat that is generated during the fermentation (in KJ)?

$$(253,000 \text{ KJ})$$

Solution:

Given:

<u>Microorganism</u>	<u>Substrate</u>	<u>Fermentation</u>
Bacteria	Glycerol – $C_3H_8O_3$	$V = 5m^3 = 5,000L$
$CH_{1.74}O_{0.43}N_{0.22}$	$MW_S = 92 \frac{gS}{moleS}$	Ammonia as nitrogen source
$MW_X = 23.7 \frac{gX}{moleX}$	$S_0 = 10 \frac{gS}{L}$	Fermentation purpose - biomass
	$S_f = 1 \frac{gS}{L}$	
	$Y_{X/S} = 0.6 \frac{gX}{gS}$	

A. $Y_{\frac{O}{S}} = ? \frac{\text{mole } O_2}{\text{mole } S}$

It is possible to solve in 2 ways:

1. Using the oxygen consumption equation (if all conditions are met).
2. Using the stoichiometry growth equation.

In the first way:

We need to find $Y_{\frac{O}{S}}$ using $Y_{\frac{O}{X}}$ and $Y_{\frac{X}{S}}$:

First, we should check if we can use the oxygen consumption equation:

- ✓ Defined carbon source? **Yes, glycerol $C_3H_8O_3$** .
- ✓ Nitrogen source is ammonia or ammonia salt? **Yes, ammonia**.
- ✓ Growth is for biomass only? **Yes, it is given in the question that the products are biomass, CO_2 and water only.**
- ✓ Known microorganism composition? **Yes, given in the question $CH_{1.74}O_{0.43}N_{0.22}$.**

All the conditions are met, so we can use the oxygen consumption equation:

$$Y_{\frac{O}{X}} = \frac{32 \cdot C + 8 \cdot H - 16 \cdot O}{Y_{\frac{X}{S}} \times Mw_S} + 0.01 \cdot O' - 0.0267 \cdot C' + 0.01714 \cdot N' - 0.08 \cdot H'$$

Given in the question: growth yield, carbon source composition and its molecular weight.

We can calculate the percentage weight of each element in the microorganism, while the microorganism molecule weight is given ($23.7 \frac{gX}{moleX}$):

$$O' = O\% = \frac{\gamma \times Mw_O}{Mw_X} \times 100\% = \frac{0.43 \times 16}{23.7} \times 100\% = 29\%$$

$$C' = C\% = \frac{\alpha \times Mw_C}{Mw_X} \times 100\% = \frac{1 \times 12}{23.7} \times 100\% = 50.6\%$$

$$N' = N\% = \frac{\delta \times Mw_N}{Mw_X} \times 100\% = \frac{0.22 \times 14}{23.7} \times 100\% = 13\%$$

$$H' = H\% = \frac{\beta \times Mw_H}{Mw_X} \times 100\% = \frac{1.74 \times 1}{23.7} \times 100\% = 7.3\%$$

We can place the numbers in the oxygen consumption equation:

$$Y_{\frac{O}{X}} = \frac{32 \cdot 3 + 8 \cdot 8 - 16 \cdot 3}{0.6 \times 92} + 0.01 \cdot 29 - 0.0267 \cdot 50.6 + 0.01714 \cdot 13 - 0.08 \cdot 7.3$$

$$Y_{\frac{O}{X}} = \frac{112}{55.2} - 1.42$$

$$Y_{\frac{O}{X}} = 0.61 \frac{gO_2}{gX}$$

The growth yield connects between $Y_{\frac{O}{X}}$ to $Y_{\frac{O}{S}}$:

$$Y_{\frac{O}{S}} = Y_{\frac{X}{S}} \times Y_{\frac{O}{X}}$$

$$Y_{\frac{O}{S}} = 0.6 \frac{gX}{gS} \times 0.61 \frac{gO_2}{gX}$$

$$Y_{\frac{O}{S}} = 0.366 \frac{gO_2}{gS}$$

In the question we've been asked to find the **molar** yield so we have to convert unit by using the molecular weights:

$$Y_{\frac{O}{S}} \left(\frac{moleO_2}{moleS} \right) = \frac{Y_{\frac{O}{S}} \left(\frac{gO_2}{gS} \right) \times MwS}{MwO_2}$$

$$Y_{\frac{O}{S}} = \frac{0.366 \frac{gO_2}{gS} \times 92 \frac{gS}{mole S}}{32 \frac{gO_2}{mole O_2}}$$

$$Y_{\frac{O}{S}} = 1.05 \frac{mole O_2}{mole S}$$

In the second way:

We can use the growth stoichiometry:



$$\text{Where: } Y_{\frac{O}{S}} = \frac{B(moleO_2)}{A(moleS)}$$

Solution steps:

1. In order to find $Y_{\frac{O}{S}}$ we need to find the coefficients of the oxygen (B) and the substrate (A).
2. We can calculate A by the growth yield given in the question.
3. In order to find B, we need to use the atoms balance:
 - a) D can be obtained directly from the microorganism formula.
 - b) Place A in the carbon balance to find P.
 - c) Place A and D in the hydrogen balance to find Q.
 - d) Place A, P and Q in the oxygen balance to find B.
4. Place A and B in the oxygen yield on the substrate to get the final answer.

the growth yield and the molecular weights of the carbon source and the microorganism are given in the question. Therefore, by the growth yield we can find A, while the equation is normalized to M=1:

$$Y_{\frac{X}{S}} = \frac{M \times Mw_X}{A \times Mw_S}$$
$$0.6 \frac{gX}{gS} = \frac{1 \text{ mole } X \times 23.7 \frac{gX}{\text{mole } X}}{A \text{ mole } S \times 92 \frac{gS}{\text{mole } S}}$$
$$A = 0.43$$

Define M=1 and balance each of the atoms:

Carbon (C): $A \times 3 = M \times 1 + P \times 1$

Hydrogen (H): $A \times 8 + D \times 3 = M \times 1.74 + Q \times 2$

Oxygen (O): $A \times 3 + B \times 2 = M \times 0.43 + P \times 2 + Q$

Nitrogen (N): $D \times 1 = M \times 0.22$

We normalized the equation to M=1, therefor, by the nitrogen balance:

$$D \times 1 = 1 \times 0.22$$
$$D = 0.22$$

Place A in the carbon (C) balance to find P:

$$0.43 \times 3 = 1 \times 1 + P \times 1$$

$$P = 0.29$$

Place A and D in the hydrogen (H) balance to find Q:

$$0.43 \times 8 + 0.22 \times 3 = 1 \times 1.74 + Q \times 2$$

$$Q = 1.18$$

Place A, P and Q in the oxygen (O) balance to find B:

$$0.43 \times 3 + B \times 2 = 1 \times 0.43 + 0.29 \times 2 + 1.18$$

$$B = 0.45$$

Now we can calculate the molar yield by using the calculated coefficients A and B:

$$Y_{\frac{O}{S}} = \frac{B \text{ mole } O_2}{A \text{ mole } S}$$

$$Y_{\frac{O}{S}} = \frac{0.45 \text{ mole } O_2}{0.43 \text{ mole } S}$$

$$Y_{\frac{O}{S}} = 1.05 \frac{\text{mole } O_2}{\text{mole } S}$$

B. $Q_{ferm} = ? KJ$

Solution rational:

1. Cooney constant correlates between the amount of external oxygen consumed to the amount of heat generated during the process.
2. The oxygen yield on the substrate correlates between the amount of consumed substrate to the amount of external oxygen consumed.
3. Since the oxygen is supplied during the whole growth process, in order to use the growth equation, we have to convert the change in the substrate concentration to the change in its amount.

We will use Cooney constant to calculate the amount of heat generated during the fermentation:

$$Q_{ferm} = Q_f \times \Delta W_{O_2}$$

$$\text{Where: } Q_f = 3.74 \frac{Kcal}{gO_2}$$

We can find the amount of oxygen consumed during the fermentation (ΔW_{O_2}) from the oxygen yield on the substrate that was calculated in section A:

$$Y_{\frac{O}{S}} = \frac{\Delta W_{O_2}}{\Delta W_S} \rightarrow \Delta W_{O_2} = Y_{\frac{O}{S}} \times \Delta W_S$$

The substrate concentration at the beginning and at the end of the fermentation are given in the question, so we can calculate the substrate weight consumed in the fermentation:

$$\begin{aligned} \Delta W_S &= (S_0 - S_f) \times V \\ \Delta W_S &= (10 - 1) \frac{gS}{L} \times 5,000L \\ \Delta W_S &= 4.5 \times 10^4 gS \end{aligned}$$

Place in the yield equation to find the consumed oxygen mass:

$$\begin{aligned} \Delta W_{O_2} &= Y_{\frac{O}{S}} \times \Delta W_S \\ \Delta W_{O_2} &= 0.366 \frac{gO_2}{gS} \times (4.5 \times 10^4) gS \\ \Delta W_{O_2} &= 16,470 gO_2 \end{aligned}$$

From here, by Cooney law, the amount of heat released during the fermentation is:

$$\begin{aligned} Q_{ferm} &= Q_f \times \Delta W_{O_2} \\ Q_{ferm} &= 3.74 \frac{Kcal}{gO_2} \times 16,470 gO_2 \\ Q_{ferm} &= 61,598 Kcal \end{aligned}$$

In the question we were asked to present the result in KJ unit, so we have to convert:

$$Q_{ferm} = 61,598 Kcal \times \frac{4.184 KJ}{1 Kcal}$$

$$Q_{ferm} = 257,725 KJ$$

Question 2

A certain protein that is essential for the industry can be produced by two methods: by bacteria growth on hexadecane or yeast growth on methanol.

Hexadecane cost is 0.35 \$/kg, while methanol cost is 0.22 \$/kg. The bacteria yield on hexadecane is 1.0 gX/gS, while the yeast yield on methanol is 0.4 gX/gS.

Assuming the protein percentage in bacteria and yeast is the same, which of the methods is more cost-effective, protein production using bacteria or yeast?

(Bacteria on hexadecane)

Solution:

Given:

<u>Yeast</u>	<u>Bacteria</u>
Methanol	Hexadecane
$Y_{\frac{X_2}{MeOH}} = 0.4 \frac{gX_2}{gM}$	$Y_{\frac{X_1}{Hex}} = 1.0 \frac{gX_1}{gH}$
$Cost_{MeOH} = 0.22 \frac{\$}{KgM}$	$Cost_{Hex} = 0.35 \frac{\$}{KgH}$

It is given in the question that the protein percentage in the yeast and bacteria is similar, so in order to find which of the processes is more cost-effective we need to calculate the production cost of 1 gram biomass in each of the processes.

For bacteria on hexadecane:

$$Y_{\frac{\$}{X_1}} = \frac{Cost_{Hex}}{Y_{\frac{X_1}{Hex}}}$$

$$Y_{\frac{\$}{X_1}} = \frac{0.35 \frac{\$}{KgH} \times \frac{1KgH}{1000gH}}{1.0 \frac{gX_1}{gH}}$$

$$Y_{\frac{\$}{X_1}} = 3.5 \times 10^{-4} \frac{\$}{gX_1}$$

For yeast on methanol:

$$Y_{\frac{\$}{X_2}} = \frac{Cost_{MeOH}}{Y_{\frac{X_1}{MeOH}}}$$

$$Y_{\frac{\$}{X_2}} = \frac{0.22 \frac{\$}{KgM} \times \frac{1KgM}{1000gM}}{0.4 \frac{gX_2}{gM}}$$

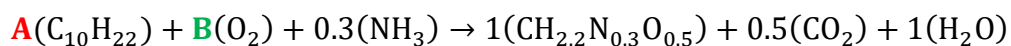
$$Y_{\frac{\$}{X_2}} = 5.5 \times 10^{-4} \frac{\$}{gX_2}$$

Protein production from bacteria growing on hexadecane is cheaper than the production using yeast on methanol.

Question 3

The bacterium *Pseudomonas* (with the empiric formula $CH_{2.2}N_{0.3}O_{0.5}$) is grown on decane ($C_{10}H_{22}$) to produce a certain protein, which constitutes 65% of the bacteria weight. During the growth, 1.25 ton of decane are utilized per 1 ton protein produced (as part of the bacteria weight).

Stoichiometry experiments show that 1.52 kg oxygen (O_2) are required for the production of 1 kg cells, and that the growth equation (partial balanced) is:



A. Calculate the values of the coefficients **A** and **B** in the growth equation.

$$(A=0.15, B=1.25)$$

B. Calculate the biomass yield on the decane ($Y_{\frac{X}{S}}$).

$$(1.23 \frac{g X}{g S})$$

C. If the ammonia gas is supplied from containers of 100 m³, how many containers will be required to produce 1 ton protein? It can be assumed that the ammonia acts as ideal gas in STP conditions.

$$(4 \text{ containers})$$

D. Assuming it is possible to convert 75% of CO₂ to dry ice, how many tons of dry ice can be produced in parallel to the production of 100 ton protein?

$$(96.8 \text{ ton})$$

Solution:

Given:

<u>Microorganism</u>	<u>Substrate</u>
Bacteria	Decane
$CH_{2.2}N_{0.3}O_{0.5}$	$C_{10}H_{22}$
$Y_{\frac{O}{X}} = 1.52 \frac{g O_2}{g X}$	$Y_{\frac{S}{P}} = 1.25 \frac{g S}{g P} = 1.25 \frac{ton S}{ton P}$
$Y_{\frac{P}{X}} = 65\% = 0.65 \frac{g P}{g X} = 0.65 \frac{ton P}{ton X}$	

A. **A**, **B** = ?

We can use the growth equation to calculate the coefficients A and B. To find A we will perform balance on the carbon atoms:

$$\text{Carbon (C): } A \times 10 = 1 \times 1 + 0.5 \times 1$$

$$A = 0.15$$

To find B we will perform balance on the oxygen atoms:

$$\text{Oxygen (O): } B \times 2 = 1 \times 0.5 + 0.5 \times 2 + 1 \times 1$$

$$B = 1.25$$

B. $Y_{\frac{X}{S}} = ?$

The biomass yield on the substrate can be calculated by using the given $Y_{\frac{P}{X}}$ and $Y_{\frac{S}{P}}$ values:

$$Y_{\frac{S}{X}} = Y_{\frac{S}{P}} \cdot Y_{\frac{P}{X}} = 1.25 \frac{g S}{g P} \cdot 0.65 \frac{g P}{g X} = 0.8125 \frac{g S}{g X}$$

$$Y_{\frac{X}{S}} = \frac{1}{Y_{\frac{S}{X}}} = \frac{1}{0.8125 \frac{g S}{g X}}$$

$$Y_{\frac{X}{S}} = 1.23 \frac{g X}{g S}$$

C. $Container_{NH_3} = ?$

Given:

$$\Delta W_P = 1 \text{ ton } P$$

First, we should find the amount of decane (the substrate) needed to produce 1 ton protein.

We can use the substrate yield on the product to calculate ΔW_S by using the given ΔW_P :

$$Y_{\frac{S}{P}} = \frac{\Delta W_S}{\Delta W_P}$$

$$1.25 \frac{g S}{g P} \cdot 10^6 \frac{g P}{\text{ton } P} = \frac{\Delta W_S}{1 \text{ ton } P}$$

$$\Delta W_S = 1.25 \text{ ton } S$$

Since the growth equation gives us the **molar** ratio between the ammonia gas to the substrate, we need to convert to the number of moles in the substrate amount that was calculated:

$$Mw_{decane} = 142 \frac{g}{\text{mole}}$$

$$n_S = \frac{1.25 \text{ ton } S}{142 \frac{g S}{\text{mole } S}} \cdot 10^6 \frac{g S}{\text{ton } S} = 8802.8 \text{ mole } S$$

Second, we need to find the number of ammonia moles consumed during the fermentation together with the amount of decane we calculated. We can use the stoichiometry ratio between decane and ammonia as detailed in the growth equation:

$$n_{NH_3} = n_S \cdot \frac{0.3}{0.15} = 17,605.6 \text{ mole } NH_3$$

It is known that ideal gas in STP conditions takes 22.4 lit/mole, therefore:

$$V_{NH_3} = n_{NH_3} \cdot 22.4 \frac{lit}{mole} = 394,366.2 lit = 394.366 m^3$$

Given in the question that the ammonia supplied from containers of 100 m³, so:

$$Container_{NH_3} = \frac{394.366 m^3}{100 m^3} = 3.94$$

4 ammonia containers (of 100 m³) are required in order to produce 1 ton protein.

D. $W_{dryice} = ?$

Given:

$$\Delta W_P = 100 \text{ ton } P$$

In a similar way as before, we first need to calculate the amount of substrate required to produce 100 ton protein. We will use the substrate yield on the product:

$$\begin{aligned} \Delta W_S &= 1.25 \frac{\text{ton } S}{\text{ton } P} \cdot 100 \text{ ton } P = 125 \text{ ton } S \\ n_S &= \frac{125 \text{ ton } S}{142 \frac{g \text{ } S}{mole \text{ } S}} \cdot 10^6 \frac{g \text{ } S}{\text{ton } S} = 880,281.7 \text{ mole } S \end{aligned}$$

From the stoichiometry ratio between the substrate and the CO₂ in the growth equation, we know that from every 0.15 mole substrate we get 0.5 mole CO₂. From here, we can calculate the weight of CO₂:

$$\begin{aligned} n_{CO_2} &= n_S \cdot \frac{0.5}{0.15} = 2934272.3 \text{ mole } CO_2 \\ Mw_{CO_2} &= 44 \frac{g \text{ } CO_2}{mole \text{ } CO_2} \\ W_{CO_2} &= 44 \frac{g \text{ } CO_2}{mole \text{ } CO_2} \cdot 10^{-6} \frac{\text{ton } CO_2}{g \text{ } CO_2} \cdot 2934272.3 \text{ mole } CO_2 = 129.1 \text{ ton } CO_2 \end{aligned}$$

Given in the question that only 75% of CO₂ can be converted to dry ice, therefore:

$$W_{dryice} = 129.1 \text{ ton } CO_2 \cdot 0.75$$

$$\boxed{W_{dryice} = 96.8 \text{ ton}}$$